Physics Worksheet

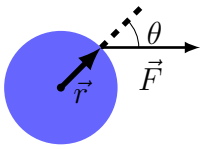
Vectors

Name: _____ **Student #:** _____ **Student Email #:** _____

1. **Algebra With Vectors**
   A boat is traveling across a river. It's going south east with a velocity of: $\vec{v_B} = (-5, -4)\text{km}\,\text{h}^{-1}$. The river is flowing with a velocity of $\vec{v_w} = (2, 1)\text{km}\,\text{h}^{-1}$. With what velocity is the boat crossing the river?

2. **Scalar Product**
   Aki and Luke go running off in different directions. $\vec{v_A} = (\frac{1}{3}, 2)\text{m}\,\text{s}^{-1}$, $\vec{v_L} = (-1, \frac{3}{4})\text{m}\,\text{s}^{-1}$. What is the angle between their velocity vectors?

3. **Cross Product**
   A force $\vec{F} = (6, 3)$ is applied to the edge of a disc centered at the origin at $\vec{r} = (1, 2)$. What is the magnitude and direction of the angular acceleration?



4. **Right Hand Rule**
   Insert questions and tikz drawings here